CS3 RESEARCH EXPERIENCE FOR TEACHERS

Introduction to Convolutional Neural Networks

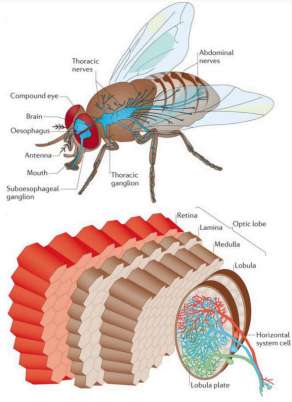
Mehmet Kerem Turkcan

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Visual systems repeat one structure everywhere

Fundamental idea

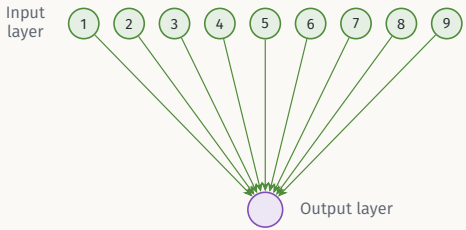
- There is a repeating structure in visual systems.
- Every patch of the input to the retina is processed in the same way.
- How do we structure our thinking about neural networks to enable this?



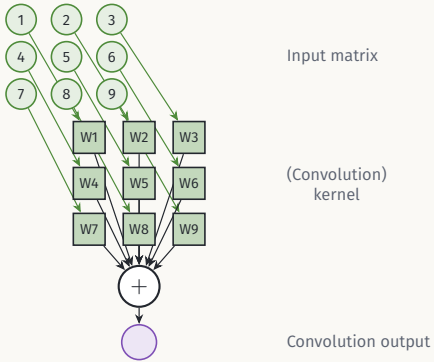
The fly nervous system (top) and the layers of its optic lobe, from retina to lobula plate (bottom).

The same unit, drawn two ways

Feedforward view of a neural network

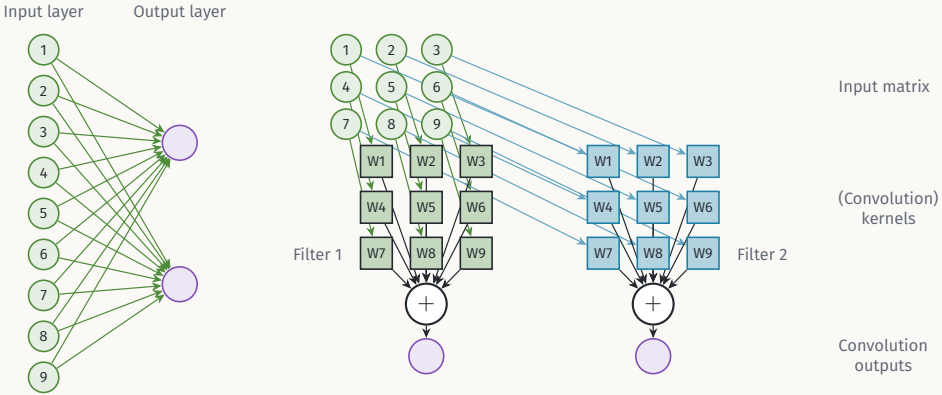


Convolutional view of a neural network



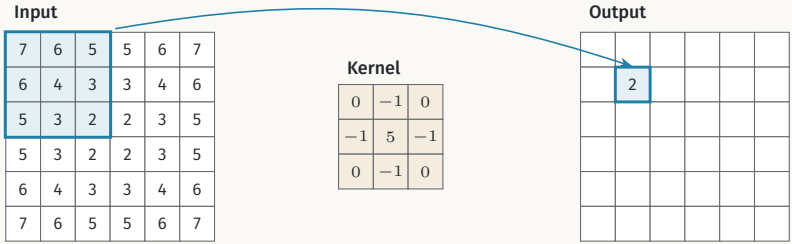
Both views compute a weighted sum of nine inputs. The convolutional view arranges the inputs as a matrix and the weights as a 3×3 **kernel**: input i is multiplied by weight W_i.

Two filters read the same input and give two outputs



Each filter has its own nine weights. Two output units in the feedforward view correspond to two kernels in the convolutional view.

Why do this? To slide the kernel over a larger input



Highlighted patch times kernel: $5 \cdot 4 - 6 - 6 - 3 - 3 = 2$

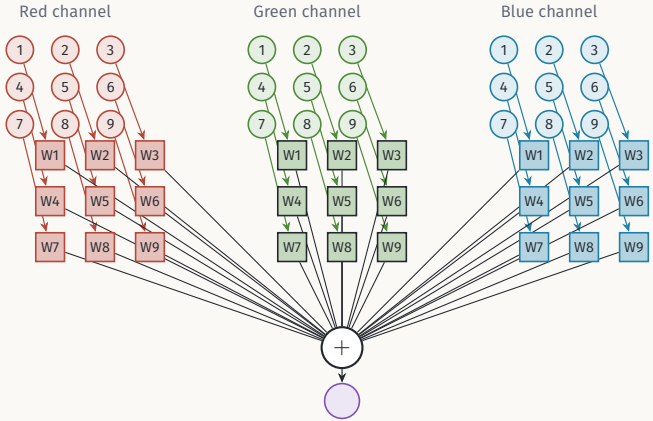
- The kernel is multiplied element by element with one 3×3 patch of the input, and the products are summed into one output value.
- Sliding the same kernel to every position fills in the whole output, so the same nine weights process every patch.

Redrawn after Michael Plotke's *2D convolution animation*, Wikimedia Commons.

Each image channel gets its own kernel

Fundamental idea

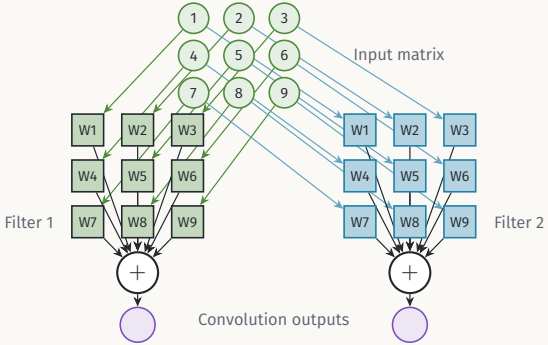
- Images have **channels** (red, green, blue).
- If a layer has K filters, it produces K output channels.
- The number of channels in a representation is also called its **depth**.



This example has 3 input channels and 1 output channel, since there is only 1 output unit.

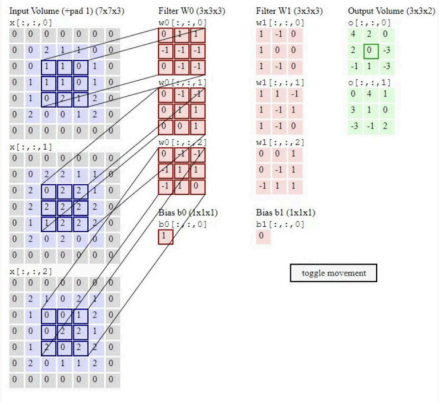
K filters produce K output channels

- One input channel feeds two filters.
- Each filter slides over the input on its own and writes its own output channel.
- With $K = 2$ filters, the output has depth 2.



This example has 1 input channel and 2 output channels, since there are 2 output units.

A convolution over a 3-channel input



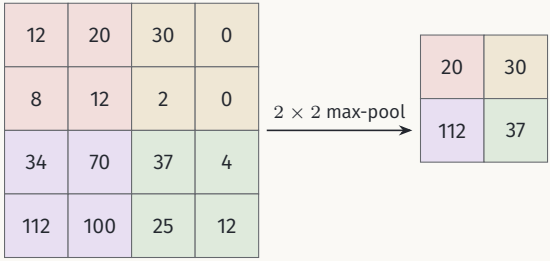
Convolution demo from Stanford CS231n, cs231n.github.io/convolutional-networks.

- **Input volume**: $7 \times 7 \times 3$, a 5×5 image with 3 channels and a border of zeros (padding 1).
- **Filters** W_0 and W_1: each is $3 \times 3 \times 3$, one 3×3 slice per input channel, plus a bias b_0 or b_1.
- **Output volume**: $3 \times 3 \times 2$, one output channel per filter.
- Each output value sums the products over all three channels, then adds the bias.
- Open the demo to watch the filters move.

Pooling shrinks the arrays between layers

Fundamental idea

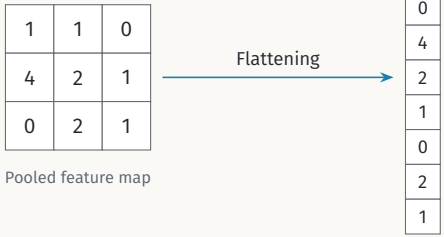
- Modern images are large: a 4K frame is 3840×2160 pixels.
- **Pooling** is a special kind of layer that reduces the size of the arrays passed from one layer to the next.
- 2×2 max-pooling keeps the largest value in each 2×2 block.



Flattening turns a feature map into a vector

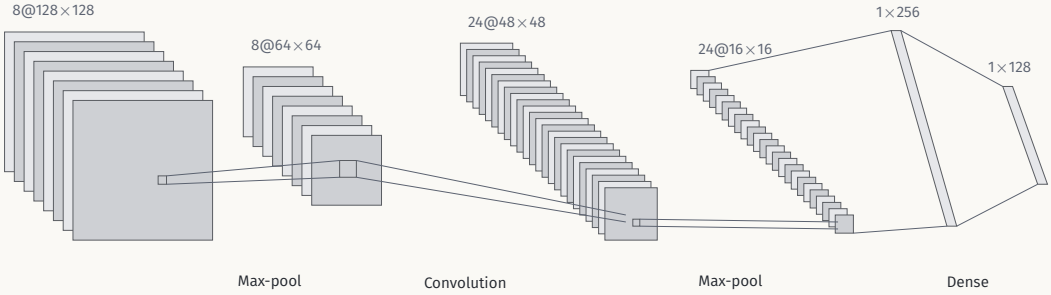
Fundamental idea

- When we classify a whole image, we no longer need its spatial structure.
- As in the strawberry and blueberry example, we can **flatten** the output of a convolutional layer.
- The network can then end like a feedforward neural network.



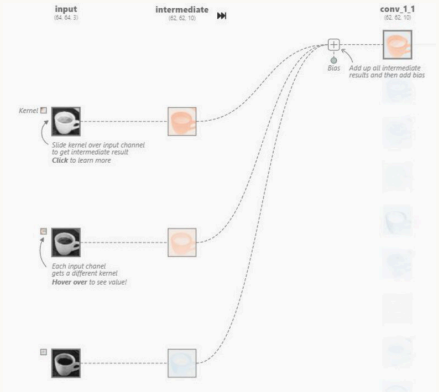
The rows of the map are read in order, one after another.

A whole convolutional neural network



Labels give channels@height×width. Convolution and pooling layers alternate; the last feature maps are flattened and passed to dense (feedforward) layers.

Explore a whole CNN interactively



Wang et al., *CNN Explainer: Learning Convolutional Neural Networks with Interactive Visualization*, IEEE VIS 2020.

- The input image has 3 channels; the first convolutional layer has 10.
- Each input channel is convolved with its own kernel, giving one intermediate result per channel.
- The intermediate results are added up, and a bias is added, to give one output channel.

Try it at poloclub.github.io/cnn-explainer.

REVIEW

Summary

- A **kernel** is a small set of shared weights that processes every patch of the input in the same way.
- Sliding the kernel over the input is a **convolution**; each position gives one output value.
- Images have **channels**; a layer with K filters produces K output channels, its **depth**.
- **Pooling** shrinks the arrays between layers, and **flattening** lets the network end like a feedforward network.